Instead of having a unified stress detection crop model, we can have a two-model system to keep the original data valid. When our physical sensors collect all the features from the same plant, both models can evaluate that observation and their predictions can be combined.

Model 1: Environmental Stress Model

This model can learn environmental stress from the large agriculture_dataset.

Features:

- Temperature
- Humidity
- Rainfall
- Soil moisture
- pH
- Wind speed
- Soil type

Output:

- 0 = Healthy
- 1 = Environmental Stress

This model detects drought, heat, excessive moisture, poor pH, and strong wind.

Model 2: Nutrient Stress Model

This model can learn nutrient stress from the same plan using the plant_health dataset.

Features:

- Nitrogen
- Phosphorus
- Potassium
- Temperature
- Humidity
- Soil moisture
- pH
- Light intensity, when available

Output:

- Healthy
- Moderate Nutrient Stress
- High Nutrient Stress

or

- 0 = Healthy
- 1 = Nutrient Stress

Combining Results:

Final risk = highest prediction from either model

For example:

Environmental stress: 85%
Nutrient stress: 25%

Final stress risk: 85%

This makes sense because a plant can be under serious environmental stress even when its nutrients are normal.

Hoeffding Adaptive Tree

The Hoeffding Adaptive Tree will be used as an online-learning model for crop-stress detection on the Raspberry Pi. It will process new sensor readings one observation at a time and predict whether the crop is healthy or stressed. Two adaptive trees will be used: one for environmental stress using variables such as temperature, humidity, rainfall, soil moisture, pH, sunlight, and wind speed, and another for nutrient stress using nitrogen, phosphorus, potassium, temperature, humidity, soil moisture, and pH.

The predictions from both models will be combined to produce an overall crop-health result and identify whether the likely cause is environmental or nutrient-related. When the actual condition of the plant is later verified, the correct label will be provided to the model so it can update itself and learn from the new observation without being retrained from the beginning. This will allow the system to gradually adapt to changes in crops, weather, seasons, soil conditions, and sensor data.